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SCOPE

The scope of this data pipeline design proposal is to establish a cloud-native data processing pipeline that seamlessly integrates AWS and Azure services. The pipeline is designed to handle incoming JSON data, transform it into a more structured format (CSV), and visualize it through AWS QuickSight. Additionally, the solution ensures effective notification mechanisms for stakeholders.

COMPONENTS

The proposed data pipeline comprises various components, each serving a specific role in the end-to-end data processing flow:

- **S3 Buckets**: Two S3 buckets, "S3 RAW AWS Inbound" and "S3 Raw Azure Inbound," act as the entry points for incoming JSON data from AWS and Azure, respectively. These buckets serve as storage for raw data.

- **AWS Lambda and Step Functions**: AWS Lambda functions play a crucial role in processing data. When data is uploaded to the "S3 RAW AWS Inbound" bucket, a notification triggers AWS Lambda, which, in turn, invokes an AWS Step Function. The Step Function orchestrates the data processing workflow.

- **Azure Notifications Hub**: In the Azure environment, the Azure Notifications Hub intercepts incoming messages and triggers an Azure Function for data processing.

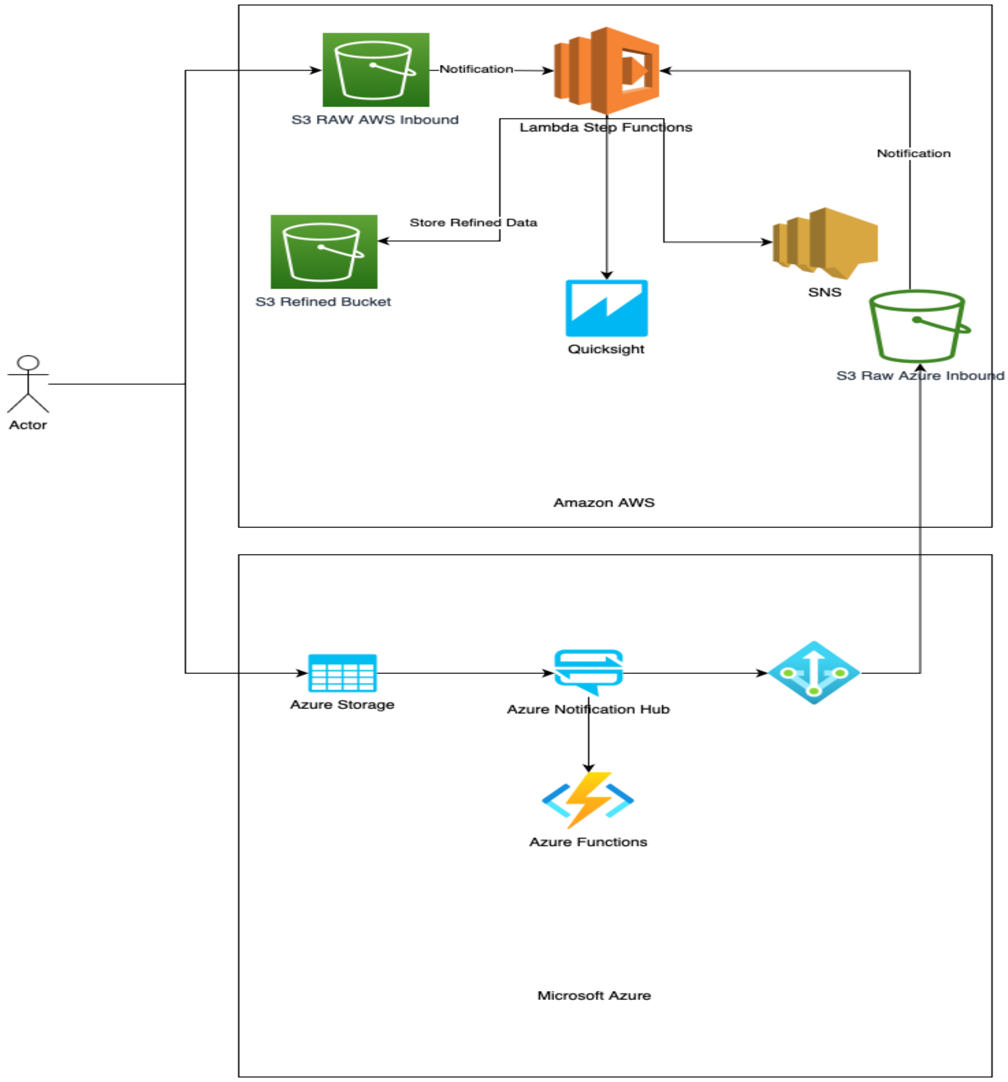
- **Data Transformation (Python Script)**: In both AWS and Azure, Python scripts within Lambda and Azure Functions are responsible for parsing JSON data and converting it into CSV format and reading the key-value pair information contained in the JSON file.

- **S3 Buckets for Refined Data**: Separate S3 buckets, "S3 Refined Bucket" for AWS and "S3 Refined Bucket" for Azure, store the processed (refined) data in CSV format.

- **AWS QuickSight**: AWS QuickSight is utilized for data visualization, allowing the creation of interactive dashboards and reports.

- **SNS Notifications**: AWS Simple Notification Service (SNS) is employed to send notifications to relevant parties, alerting them of completed data processing and report availability.

DESIGN DIAGRAM



STEPS

The data pipeline follows a sequence of steps to process and visualize data:

Data Ingestion:

Inbound JSON data is uploaded to the "S3 RAW AWS Inbound" and "S3 Raw Azure Inbound" buckets.

AWS Data Processing and Transformation:

- In AWS, a notification from "S3 RAW AWS Inbound" triggers an AWS Lambda function.
- The Lambda function invokes an AWS Step Function.
- The Step Function initiates data processing and transformation, converting it from JSON to CSV format.
- The resulting CSV is placed in the "S3 Refined Bucket" (AWS).
- Another step pushes the CSV to AWS QuickSight.
- A final step triggers an SNS notification.

Azure Data Processing:

- In Azure, the Azure Notifications Hub intercepts inbound messages.
- This hub triggers an Azure Function for JSON to CSV transformation.
- The CSV data is placed in "S3 Raw Azure Inbound" (AWS).
- A notification flow is initiated for AWS Lambda Step Function.
- The AWS Lambda Step Function follows the same sequence as in the AWS data processing steps.

CONCLUSION

The proposed cloud-native data pipeline effectively addresses the requirements of processing and visualizing JSON data from both AWS and Azure sources. By leveraging AWS and Azure services in a coordinated manner, the solution ensures data integrity, scalability, and seamless integration between cloud environments. It embraces event-driven architectures, enhances data processing efficiency, and provides timely notifications to stakeholders, contributing to a robust and reliable data pipeline.

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